WORMPROPAGATION FOR TRAINING THE DLCOSMOS
[ALI NAJIB] CORE FILE: CREATION: CONSTRUCTION-CYCLE 2 PRIME PRIME PRIME PRIME

Creation. DLCOSMOS, WormPropagation. Wormpropagation, in key words. DLCosmos Wormpropagation. The Worm is a Brownian particle. Try all permutations of the worm, note that loops give circularity, ∞-length worm gives probable WormPropagation convergence and lengthy derivatives, but takes long to train, so consider worms of length $< \infty$, don't loop to not lose the cross-ness, still need convergence, hyperparametric trainer, or loop and have sure convergence. Wormhole WormPropagation works, for any exotic NN, so higher NN-interlink-order too!!! This was not so in priorBackPropagation. This is the most elegent non-precise computational solution, and herein lay the infinite derivaties. Find one worm through [HyperparmetricTrainerDRL], one hyperparameter is worm-length, which can be of level ∞, $\infty-$length worm gives probable convergence and lengthy derivatives, BPTT is accounted for. There also an executor DRL, [HyperparametricExecutorDRL], loops are possible here, not during training! $[\text{CrossCNNRNNDLCosmos}]^s_{\mathcal{D}_f}$, train according to worm-directive, if worm goes through graph as

$$A \to B \to C \to B \to C \to A$$

Backpropagate as

$$A \leftarrow B \leftarrow C \leftarrow B \leftarrow C \leftarrow A$$

This is to maintain the chain $f_1 \circ f_2 \circ f_3 \circ f_4 \circ f_5$. Now we just need convergence through ∞, so let all go well through [HyperparametricConvergenceDRL] included [HyperparametricDRL] to let all go well in Wormpropagator. Finite, loopless wormpropagation is sampking neural networks, this is one way to account for crossness. We can now predict financial markets and everything. Inifnite loopless converges, finite loopless needs [HyperparametricDRL], finite looped converges. Most interesting case of mulitple worms deplyed is possible. Looped propagation gives circular DL upon worm deployment. Loopless propagation includes looped propagation and gives first non-circular worm-deployment. Convergence can be automatic or if necessary through [HyperparametricDRL]. This might a result of discrete stochastic processes convergence would be remarkable especially through multi-processing multi-wormpropagation.

DLCOSMOS. Exact parameter calculation, with WormPropagation as constant back-up for when things go wrong. We consider a $[\text{DLStarSystem}]_{\mathcal{D}_f}$ with 3 nodes and interlink-order 1.

$$\begin{aligned}
\mathbf{x}_A &= \mathbf{x}_A(\mathbf{x}_B, \mathbf{x}_C, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F, \mathbf{x}_G, \mathbf{M}) \\
\mathbf{x}_B &= \mathbf{x}_B(\mathbf{x}_A, \mathbf{x}_C, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F, \mathbf{M}) \\
\mathbf{x}_C &= \mathbf{x}_C(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F, \mathbf{x}_H, \mathbf{M}) \\
\mathbf{x}_D &= \mathbf{x}_B(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C, \mathbf{x}_E, \mathbf{x}_F, \mathbf{M}) \\
\mathbf{x}_E &= \mathbf{x}_E(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C, \mathbf{x}_D, \mathbf{x}_F, \mathbf{x}_I, \mathbf{M}) \\
\mathbf{x}_G &= \mathbf{x}_G(\mathbf{x}_A, \mathbf{M}) \\
\mathbf{x}_H &= \mathbf{x}_H(\mathbf{x}_C, \mathbf{M}) \\
\mathbf{x}_I &= \mathbf{x}_I(\mathbf{x}_E, \mathbf{M})
\end{aligned}$$

Where

$$\begin{aligned}\mathbf{M} = (&\mathbf{M}_{AB}, \mathbf{M}_{AC}, \mathbf{M}_{AD}, \mathbf{M}_{AE}, \mathbf{M}_{AF}, \mathbf{M}_{AG},\\ &\mathbf{M}_{BA}, \mathbf{M}_{BC}, \mathbf{M}_{BD}, \mathbf{M}_{BE}, \mathbf{M}_{BF},\\ &\mathbf{M}_{CA}, \mathbf{M}_{CB}, \mathbf{M}_{CD}, \mathbf{M}_{CE}, \mathbf{M}_{CF}, \mathbf{M}_{CH},\\ &\mathbf{M}_{DA}, \mathbf{M}_{DB}, \mathbf{M}_{DC}, \mathbf{M}_{DE}, \mathbf{M}_{DF},\\ &\mathbf{M}_{EA}, \mathbf{M}_{EB}, \mathbf{M}_{EC}, \mathbf{M}_{ED}, \mathbf{M}_{EF}, \mathbf{M}_{EI},\\ &\mathbf{M}_{GA}, \mathbf{M}_{HC}, \mathbf{M}_{IE})\end{aligned}$$

Now abbreviate e.g. $\mathbf{x}_A(\mathbf{x}_B, \mathbf{x}_C, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F, \mathbf{x}_G, \mathbf{M})$ by $\mathbf{x}_A(\mathbf{M})$, since only derivates w.r.t. to parameters are needed. The full equations now read as

$$\begin{aligned}\mathbf{x}_A(M) &= \sigma(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M}))\\ \mathbf{x}_B(M) &= \sigma(\mathbf{M}_{AB}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_G(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M}))\\ \mathbf{x}_C(M) &= \sigma(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M}))\\ \mathbf{x}_D(M) &= \sigma(\mathbf{M}_{AD}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M}))\\ \mathbf{x}_E(M) &= \sigma(\mathbf{M}_{AE}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M}))\\ \mathbf{x}_F(M) &= \sigma(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M}))\\ \mathbf{x}_G(\mathbf{M}) &= \sigma(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M}))\\ \mathbf{x}_H(\mathbf{M}) &= \sigma(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M}))\\ \mathbf{x}_I(\mathbf{M}) &= \sigma(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M}))\end{aligned}$$

Now take the derivative w.r.t. all of $\mathbf{M}$, and recall the composition of $\mathbf{M}$, so start with taking the derivative w.r.t. $\mathbf{M}_{BA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}})\\ &\quad \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BA}}\\ &\quad \oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})\\ &\quad \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}}\\ &\quad \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatices. So continue by taking the derivative w.r.t. $\mathbf{M}_{CA}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}}$$
$$\oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}}$$
$$\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatices. So continue by taking the derivative w.r.t. $\mathbf{M}_{DA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DA}}) \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DA}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatices. So continue by taking the derivative w.r.t. $\mathbf{M}_{DA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \oplus ([\mathbf{x}_G(\mathbf{M})] + \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}}) \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{GA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{GA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EA}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EA}} \\
&\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} \\
&\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \\
&\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EA}} \\
&\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} \\
&\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \\
&\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{FA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}}))\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AB}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot ((\mathbf{x}_A(\mathbf{M}) + \mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AB}}$$
$$\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AB}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AB}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CB}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus (\mathbf{x}_C(\mathbf{M}) + \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}}) \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CB}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DB}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus (\mathbf{x}_D(\mathbf{M}) + \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CD}}) \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DB}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DB}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EB}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} \\
&\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} \\
&\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \\
&\oplus (\mathbf{x}_E(\mathbf{M}) + \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} \\
&\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} \\
&\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \\
&\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FB}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus (\mathbf{x}_F(\mathbf{M}) + \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}}))\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$
$$\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AC}}$$
$$\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BC}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \\ &\oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BC}}) \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DC}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}}) \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}}) \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DC}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{HC}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}} \\
&\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{HC}} \\
&\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \\
&\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \\
&\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}}) \oplus ([\mathbf{x}_H(\mathbf{M})] + \mathbf{M}_{HC}\frac{\mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DC}}) \\
&\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \\
&\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{HC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{HC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{HC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EC}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}}) \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}}))$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}}$$
$$\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FC}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FC}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FC}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AD}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}}) \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AD}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BD}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BD}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BD}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BD}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CD}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\
&\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\
&\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\
&\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\
&\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\
&\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}}) \\
&\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CD}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CD}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CD}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CD}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{ED}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \\ &\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{ED}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{ED}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{ED}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FD}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{FD}}))$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}}$$
$$\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FD}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FD}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FD}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{AE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}}) \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{BE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CE}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{CE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\
&\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CE}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{IE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{IE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \oplus ([\mathbf{x}_I(\mathbf{M})] + \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{IE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{IE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{IE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DE}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}}$$
$$\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}}))\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FE}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{AF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AF}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AF}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BF}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{BF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\
&\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BF}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BF}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BF}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{CF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}}) \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CF}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DF}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} = \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} = \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} = \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} = \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{DF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} = \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \\ &\oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}}) \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DF}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DF}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ &\oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}}) \oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}}))\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AG}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EF}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot ([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AG}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AG}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}})
\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CH}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})
\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})
\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})
\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\
&\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CH}})
\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} &= \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}} &= \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{CH}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{CH}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \oplus \mathbf{M}_{CH}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CH}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CH}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CH}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EI}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} &= \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} &= \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EI}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}}) \oplus \mathbf{M}_{AG}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EI}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EI}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EI}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

Here are the derivatives w.r.t. $\mathbf{M}_{CA}$

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \\ &\oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AB}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}} = \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CA}} &= \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

The following enumeration defines the algorithm InnerConvergence($\mathbf{x}, \mathbf{M}$). Repeat the iterating over the below enumeration until convergence in all $\frac{\partial \mathbf{x}}{\partial \mathbf{M}}$ reached.

1. Start by assigning $\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}}$ its calculated derivative.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}}) \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

2. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

3. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

4. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

5. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

6. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

7. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BA}})\end{aligned}$$

8. Continue by assigning $\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}}$ its derivative.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \\ &\oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

9. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

10. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

11. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

12. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

13. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BA}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})\end{aligned}$$

14. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}
\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\
\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CA}}) \\
\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CA}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CA}})
\end{aligned}$$

15. Use the structure in the equations for ease in taking derivatices. So continue by taking the derivative w.r.t. $\mathbf{M}_{DA}$.

$$\begin{aligned}
\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\
&\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \\
&\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \oplus ([\mathbf{x}_G(\mathbf{M})] + \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}}) \\
&\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})
\end{aligned}$$

16. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \\
&\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})
\end{aligned}$$

17. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\
&\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \\
&\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}} \\
&\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})
\end{aligned}$$

18. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}
\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\
&\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{GA}} \\
&\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})
\end{aligned}$$

19. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

20. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{GA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{GA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

21. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{GA}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{GA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{GA}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{GA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{GA}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{GA}})\end{aligned}$$

22. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

23. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{GA}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

24. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

25. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

26. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

27. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EA}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EA}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

28. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EA}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EA}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EA}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EA}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EA}})\end{aligned}$$

29. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FA}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{FA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}}))\end{aligned}$$

30. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

31. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

32. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

33. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FA}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}})\end{aligned}$$

34. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FA}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FA}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FA}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}})$$

35. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FA}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FA}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FA}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FA}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FA}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FA}})$$

36. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AB}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

37. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot ((\mathbf{x}_A(\mathbf{M}) + \mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

38. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})$$

39. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

40. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

41. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AB}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

42. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AB}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AB}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AB}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AB}})\end{aligned}$$

43. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CB}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})\end{aligned}$$

44. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus (\mathbf{x}_C(\mathbf{M}) + \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}}) \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

45. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

46. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

47. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

48. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{CB}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

49. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CB}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CB}})$$

50. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DB}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

51. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus (\mathbf{x}_D(\mathbf{M}) + \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{CD}}) \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

52. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

53. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})\end{aligned}$$

54. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}})$$

55. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DB}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DB}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}})$$

56. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DB}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DB}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DB}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DB}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DB}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DB}})$$

57. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EB}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})$$

58. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \oplus (\mathbf{x}_E(\mathbf{M}) + \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})$$

59. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

60. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

61. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

62. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EB}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

63. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EB}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EB}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EB}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EB}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}})\end{aligned}$$

Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FB}$.

64.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

65. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus (\mathbf{x}_F(\mathbf{M}) + \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}}))$$

66. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{BC}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

67. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

68. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EB}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M})$$
$$\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FB}}$$
$$\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

69. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{FB}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FB}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

70. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FB}})$$

71. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

72. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

73. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}}) \oplus \mathbf{M}_{BC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

74. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

75. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

76. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{AC}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

77. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AC}})$$

78. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

79. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FB}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

80. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BC}}) \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

81. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

82. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

83. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{BC}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

84. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BC}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BC}})$$

85. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DC}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

86. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

87. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AC}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DC}}) \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}}) \oplus \mathbf{M}_{HC}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

88. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{DC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})\end{aligned}$$

89. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DC}} \\ \oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}})$$

90. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DC}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{DC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{DC}} \\ \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}})$$

91. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DC}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DC}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DC}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DC}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DC}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DC}})$$

92. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{HC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}} \\ \oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{HC}} \\ \oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})$$

93. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \\ \oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})$$

94. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}}) \oplus ([\mathbf{x}_H(\mathbf{M})] + \mathbf{M}_{HC}\frac{\mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DC}}) \\ &\oplus \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

95. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

96. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

97. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{HC}} \\ &\oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

98. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{HC}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{HC}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{HC}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{HC}})\end{aligned}$$

99. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EC}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{CA}\frac{\mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DA}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{GA}\frac{\mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{EA}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FA}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

100. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{CB}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{DB}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{EB}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FB}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

101. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{DC}\frac{\mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DC}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{HC}} \oplus \mathbf{M}_{EC}\frac{\mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EC}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}}) \oplus \mathbf{M}_{FC}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

102. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BD}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{CD}\frac{\mathbf{x}_D(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{ED}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FD}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

103. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BE}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EC}} \\ &\oplus \mathbf{M}_{CE}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{IE}\frac{\mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DE}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{FE}\frac{\mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}})\end{aligned}$$

104. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BF}\frac{\mathbf{x}_B(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{CF}\frac{\mathbf{x}_C(\mathbf{M})}{\mathbf{M}_{EC}} \oplus \mathbf{M}_{DF}\frac{\mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{EF}\frac{\mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$

105. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$

106. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FC}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

107. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

108. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{HC}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}}))$$

109. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EC}})$$

110. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

111. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EC}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

112. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FC}})$$

113. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AD}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

114. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

115. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

116. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FC}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}}) \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

117. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

118. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}} \\ \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

119. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AD}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AD}})$$

120. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BD}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

121. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

122. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

123. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M}}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

124. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

125. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BD}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

126. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BD}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BD}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BD}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BD}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BD}})\end{aligned}$$

127. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CD}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

128. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

129. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

130. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

131. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

132. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CD}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CD}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CD}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CD}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CD}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CD}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CD}})\end{aligned}$$

133. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{ED}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

134. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

135. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

136. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \\ &\oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}}) \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

137. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}})\end{aligned}$$

138. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{ED}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}})$$

139. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{ED}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{ED}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{ED}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{ED}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{ED}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{ED}})$$

140. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FD}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

141. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

142. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

143. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \\ \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{FD}}))$$

144. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \\ \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

145. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}} \\ \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

146. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FD}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FD}})$$

147. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AE}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AE}} \\ \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

148. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

149. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

150. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{AE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

151. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}}) \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

152. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

153. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AE}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AE}})$$

154. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

155. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

156. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

157. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{BE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

158. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

159. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BE}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

160. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BE}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BE}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BE}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BE}})\end{aligned}$$

161. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

162. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

163. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

164. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{CE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

165. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

166. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CE}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

167. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CE}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CE}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CE}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CE}})\end{aligned}$$

168. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{IE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

169. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

170. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

171. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{IE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{IE}})\end{aligned}$$

172. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \oplus ([\mathbf{x}_I(\mathbf{M})] + \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{IE}}) \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M}}{\partial \mathbf{M}_{IE}}\end{aligned}$$

173. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{IE}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}})$$

174. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{IE}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{IE}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{IE}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{IE}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{IE}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{IE}})$$

175. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DE}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

176. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

177. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}})$$

178. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{DE}})\end{aligned}$$

179. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}}\end{aligned}$$

180. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DE}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}})\end{aligned}$$

181. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DE}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DE}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DE}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DE}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DE}})\end{aligned}$$

182. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{FE}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})\end{aligned}$$

183. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$

184. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$

185. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$

186. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus ([\mathbf{x}_F(\mathbf{M})] + \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}}))$$

187. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}} \\ \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{FE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$

188. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{FE}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{FE}})$$

189. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

190. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

191. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

192. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{AF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

193. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

194. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AF}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

195. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{AF}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AF}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AF}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AF}})\end{aligned}$$

196. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{BF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

197. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

198. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

199. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{BF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

200. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

201. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{BF}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus ([\mathbf{x}_B(\mathbf{M})] + \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}} \\ &\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

202. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{BF}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{BF}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{BF}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{BF}})\end{aligned}$$

203. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CF}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

204. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

205. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

206. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{CF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

207. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}})\end{aligned}$$

208. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CF}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M})$$
$$\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}})$$
$$\oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}})$$

209. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CF}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CF}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CF}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CF}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CF}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CF}})$$

210. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{DF}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M})$$
$$\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

211. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M})$$
$$\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

212. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M})$$
$$\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{DF}}$$
$$\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

213. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ \oplus \mathbf{M}_{DF}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \\ \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

214. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \\ \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

215. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}} \\ \oplus ([\mathbf{x}_D(\mathbf{M})] + \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{DF}}) \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

216. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{DF}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{DF}})$$

217. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EF}$.

$$\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ \oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ \oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{EF}} \\ \oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

218. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

219. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

220. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

221. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

222. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EF}}) \oplus ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}}))$$

223. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EF}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EF}})$$

224. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{AG}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

225. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

226. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

227. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

228. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

229. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

230. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EF}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot ([\mathbf{x}_A(\mathbf{M})] + \mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{AG}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{AG}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{AG}})\end{aligned}$$

231. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{CH}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

232. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

233. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

234. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

235. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

236. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{CH}} \leftarrow \sigma'(&\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \\ &\oplus \mathbf{M}_{CH}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}} \\ &\oplus \mathbf{M}_{CH}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \oplus \mathbf{M}_{CH}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

237. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\begin{aligned}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{CH}} &\leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \\ \frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{CH}} &\leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot ([\mathbf{x}_C(\mathbf{M})] + \mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{CH}}) \\ \frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{CH}} &\leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{CH}})\end{aligned}$$

238. Use the structure in the equations for ease in taking derivatives. So continue by taking the derivative w.r.t. $\mathbf{M}_{EI}$.

$$\begin{aligned}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}} \leftarrow \sigma'(&\mathbf{M}_{BA}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CA}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DA}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{GA}\mathbf{x}_G(\mathbf{M}) \\ &\oplus \mathbf{M}_{EA}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FA}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{BA}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{CA}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DA}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{GA}\frac{\partial \mathbf{x}_G(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EA}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FA}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

239. For $\mathbf{x}_B(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \leftarrow \sigma'(&\mathbf{M}_{AB}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{CB}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DB}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EB}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FB}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AB}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CB}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DB}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EB}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FB}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

240. For $\mathbf{x}_C(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \leftarrow \sigma'(&\mathbf{M}_{AC}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BC}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{DC}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{HC}\mathbf{x}_H(\mathbf{M}) \\ &\oplus \mathbf{M}_{EC}\mathbf{x}_E(\mathbf{M}) \oplus \mathbf{M}_{FC}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AC}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DC}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{HC}\frac{\partial \mathbf{x}_H(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{EC}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FC}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

241. For $\mathbf{x}_D(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \leftarrow \sigma'(&\mathbf{M}_{AD}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{BD}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{CD}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{ED}\mathbf{x}_E(\mathbf{M}) \\ &\oplus \mathbf{M}_{FD}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AD}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BD}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CD}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{ED}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FD}\frac{\partial \mathbf{x}_{FC}(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

242. For $\mathbf{x}_E(\mathbf{M})$, this amounts to

$$\begin{aligned}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}} \leftarrow \sigma'(&\mathbf{M}_{AE}\mathbf{x}_{BA}(\mathbf{M}) \oplus \mathbf{M}_{BE}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CE}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{IE}\mathbf{x}_I(\mathbf{M}) \\ &\oplus \mathbf{M}_{DE}\mathbf{x}_D(\mathbf{M})) \oplus \mathbf{M}_{FE}\mathbf{x}_F(\mathbf{M})) \cdot (\mathbf{M}_{AE}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BE}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \\ &\oplus \mathbf{M}_{CE}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{IE}\frac{\partial \mathbf{x}_I(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DE}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{FE}\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}})\end{aligned}$$

243. For $\mathbf{x}_F(\mathbf{M})$, this amounts to

$$\frac{\partial \mathbf{x}_F(\mathbf{M})}{\partial \mathbf{M}_{EI}} \leftarrow \sigma'(\mathbf{M}_{AF}\mathbf{x}_A(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B(\mathbf{M}) \oplus \mathbf{M}_{CF}\mathbf{x}_C(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D(\mathbf{M}) \oplus \mathbf{M}_{EF}\mathbf{x}_E(\mathbf{M})) \cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{EI}\frac{\partial \mathbf{x}_D(\mathbf{M})}{\partial \mathbf{M}_{EI}}) \oplus \mathbf{M}_{AG}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}})$$

244. The three terminal points $\mathbf{x}_G(\mathbf{M}), \mathbf{x}_H(\mathbf{M}), \mathbf{x}_I(\mathbf{M})$ have the following derivatives

$$\frac{\partial \mathbf{x}_G(\mathbf{M})}{\mathbf{M}_{EI}} \leftarrow \sigma'(\mathbf{M}_{AG}\mathbf{x}_A(\mathbf{M})) \cdot (\mathbf{M}_{AG}\frac{\partial \mathbf{x}_A(\mathbf{M})}{\partial \mathbf{M}_{AG}})$$
$$\frac{\partial \mathbf{x}_H(\mathbf{M})}{\mathbf{M}_{EI}} \leftarrow \sigma'(\mathbf{M}_{CH}\mathbf{x}_C(\mathbf{M})) \cdot (\mathbf{M}_{CH}\frac{\partial \mathbf{x}_C(\mathbf{M})}{\partial \mathbf{M}_{AG}})$$
$$\frac{\partial \mathbf{x}_I(\mathbf{M})}{\mathbf{M}_{EI}} \leftarrow \sigma'(\mathbf{M}_{EI}\mathbf{x}_E(\mathbf{M})) \cdot ([\mathbf{x}_E(\mathbf{M})] + \mathbf{M}_{EI}\frac{\partial \mathbf{x}_E(\mathbf{M})}{\partial \mathbf{M}_{EI}})$$

Now consider SGD, and the loss-derivative $\nabla_{\mathbf{M}} Q^F(\mathbf{M})$ that comes in it. One derivative reads as follows.

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F|| \\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||' \\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) \\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\mathbf{M}_{ED}})\end{aligned}$$

Note that this derivative is now easily calculated. Now take the derivates w.r.t. to all of the below matrices. Note that these derivative are easily calculated too.

$$\begin{aligned}\mathbf{M} = (&\mathbf{M}_{AB}, \mathbf{M}_{AC}, \mathbf{M}_{AD}, \mathbf{M}_{AE}, \mathbf{M}_{AF}, \mathbf{M}_{AG}, \\ &\mathbf{M}_{BA}, \mathbf{M}_{BC}, \mathbf{M}_{BD}, \mathbf{M}_{BE}, \mathbf{M}_{BF}, \\ &\mathbf{M}_{CA}, \mathbf{M}_{CB}, \mathbf{M}_{CD}, \mathbf{M}_{CE}, \mathbf{M}_{CF}, \mathbf{M}_{CH}, \\ &\mathbf{M}_{DA}, \mathbf{M}_{DB}, \mathbf{M}_{DC}, \mathbf{M}_{DE}, \mathbf{M}_{DF}, \\ &\mathbf{M}_{EA}, \mathbf{M}_{EB}, \mathbf{M}_{EC}, \mathbf{M}_{ED}, \mathbf{M}_{EF}, \mathbf{M}_{EI}, \\ &\mathbf{M}_{GA}, \mathbf{M}_{HC}, \mathbf{M}_{IE})\end{aligned}$$

The algorithm that follows from calculating these derivatives reads as follows.

1. The derivative w.r.t. $\mathbf{M}_{ED}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F|| \\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||' \\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) \\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\mathbf{M}_{ED}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\mathbf{M}_{ED}}\end{aligned}$$

2. The derivative w.r.t. $\mathbf{M}_{AB}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AB}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\mathbf{M}_{AB}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\mathbf{M}_{A}}\end{aligned}$$

3. The derivative w.r.t. $\mathbf{M}_{AC}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AC}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial\mathbf{M}_{AC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial\mathbf{M}_{AC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial\mathbf{M}_{AC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial\mathbf{M}_{AC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial\mathbf{M}_{A}}\end{aligned}$$

4. The derivative w.r.t. $\mathbf{M}_{AD}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AD}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial\mathbf{M}_{AD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial\mathbf{M}_{AD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial\mathbf{M}_{AD}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial\mathbf{M}_{AD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial\mathbf{M}_{A}}\end{aligned}$$

5. The derivative w.r.t. $\mathbf{M}_{AE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AE}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial\mathbf{M}_{AE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial\mathbf{M}_{AE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial\mathbf{M}_{AE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial\mathbf{M}_{AE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial\mathbf{M}_{A}}\end{aligned}$$

6. The derivative w.r.t. $\mathbf{M}_{AF}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AF}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AE}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot ((\mathbf{x}_A^i] + \mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial\mathbf{M}_{AF}}) \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial\mathbf{M}_{AF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial\mathbf{M}_{AF}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial\mathbf{M}_{AF}} \oplus \mathbf{M}\end{aligned}$$

7. The derivative w.r.t. $\mathbf{M}_{AE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{AG}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{AG}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{AG}}\end{aligned}$$

8. The derivative w.r.t. $\mathbf{M}_{BA}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{BA}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{BA}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{BA}}\end{aligned}$$

9. The derivative w.r.t. $\mathbf{M}_{BC}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{BC}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{BC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{BC}}\end{aligned}$$

10. The derivative w.r.t. $\mathbf{M}_{BD}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{BD}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{BD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{BD}}\end{aligned}$$

11. The derivative w.r.t. $\mathbf{M}_{BE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{BE}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{BE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{BE}}\end{aligned}$$

12. The derivative w.r.t. $\mathbf{M}_{BF}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{BF}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus ([\mathbf{x}_B^i] + \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{BF}}) \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{BF}} \oplus \mathbf{M}\end{aligned}$$

13. The derivative w.r.t. $\mathbf{M}_{CA}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CA}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CA}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial \mathbf{M}_{C}}\end{aligned}$$

14. The derivative w.r.t. $\mathbf{M}_{CB}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CB}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CB}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial \mathbf{M}_{C}}\end{aligned}$$

15. The derivative w.r.t. $\mathbf{M}_{CD}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CD}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CD}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial \mathbf{M}_{C}}\end{aligned}$$

16. The derivative w.r.t. $\mathbf{M}_{CE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CE}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial \mathbf{M}_{C}}\end{aligned}$$

17. The derivative w.r.t. $\mathbf{M}_{CF}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CF}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus ([\mathbf{x}_C^i] + \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CF}}) \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CF}} \oplus \mathbf{M}_{EF}\end{aligned}$$

18. The derivative w.r.t. $\mathbf{M}_{CH}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{CH}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{CH}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{CH}}\end{aligned}$$

19. The derivative w.r.t. $\mathbf{M}_{DA}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{DA}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{DA}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{DA}}\end{aligned}$$

20. The derivative w.r.t. $\mathbf{M}_{DB}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{DB}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{DB}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{DB}}\end{aligned}$$

21. The derivative w.r.t. $\mathbf{M}_{DC}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{DC}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{DC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{DC}}\end{aligned}$$

22. The derivative w.r.t. $\mathbf{M}_{DE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{DE}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{DE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(}{\partial \mathbf{M}_{}}\end{aligned}$$

23. The derivative w.r.t. $\mathbf{M}_{DF}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{DF}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{DF}} \oplus ([\mathbf{x}_D^i] + \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{DF}}) \oplus \mathbf{M}\end{aligned}$$

24. The derivative w.r.t. $\mathbf{M}_{EA}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{EA}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{EA}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(}{\partial \mathbf{M}_{}}\end{aligned}$$

25. The derivative w.r.t. $\mathbf{M}_{EB}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{EB}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{EB}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(}{\partial \mathbf{M}_{}}\end{aligned}$$

26. The derivative w.r.t. $\mathbf{M}_{EC}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{EC}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{EC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(}{\partial \mathbf{M}_{}}\end{aligned}$$

27. The derivative w.r.t. $\mathbf{M}_{ED}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{ED}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{ED}}\end{aligned}$$

28. The derivative w.r.t. $\mathbf{M}_{DF}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{EF}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{EF}} \oplus ([\mathbf{x}_E^i] + \mathbf{M}_{EF}\end{aligned}$$

29. The derivative w.r.t. $\mathbf{M}_{EI}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{EI}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{EI}}\end{aligned}$$

30. The derivative w.r.t. $\mathbf{M}_{GA}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{GA}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{GA}}\end{aligned}$$

31. The derivative w.r.t. $\mathbf{M}_{HC}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{ED}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{HC}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M})}{\partial \mathbf{M}_{HC}}\end{aligned}$$

32. The derivative w.r.t. $\mathbf{M}_{IE}$ reads as follows

$$\begin{aligned}\frac{\partial \mathbf{Q}^F}{\partial \mathbf{M}_{IE}} = 2\sum_{i=1}^{n} &||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||\\ &\cdot ||\sigma(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M})) - \mathbf{x}_F||'\\ &\sigma'(\mathbf{M}_{AF}\mathbf{x}_A^i(\mathbf{M}) \oplus \mathbf{M}_{BF}\mathbf{x}_B^i \oplus \mathbf{M}_{CF}\mathbf{x}_C^i(\mathbf{M}) \oplus \mathbf{M}_{DF}\mathbf{x}_D^i \oplus \mathbf{M}_{EF}\mathbf{x}^i(\mathbf{M}))\\ &\cdot (\mathbf{M}_{AF}\frac{\partial \mathbf{x}_A^i(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{BF}\frac{\partial \mathbf{x}_B^i(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{CF}\frac{\partial \mathbf{x}_C^i(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{DF}\frac{\partial \mathbf{x}_D^i(\mathbf{M})}{\partial \mathbf{M}_{IE}} \oplus \mathbf{M}_{EF}\frac{\partial \mathbf{x}_E^i(\mathbf{M}}{\partial \mathbf{M}_{I}}\end{aligned}$$

In a similar way can all the derivatives w.r.t.

$$\begin{aligned}\mathbf{M} = (&\mathbf{M}_{AB}, \mathbf{M}_{AC}, \mathbf{M}_{AD}, \mathbf{M}_{AE}, \mathbf{M}_{AF}, \mathbf{M}_{AG},\\ &\mathbf{M}_{BA}, \mathbf{M}_{BC}, \mathbf{M}_{BD}, \mathbf{M}_{BE}, \mathbf{M}_{BF},\\ &\mathbf{M}_{CA}, \mathbf{M}_{CB}, \mathbf{M}_{CD}, \mathbf{M}_{CE}, \mathbf{M}_{CF}, \mathbf{M}_{CH},\\ &\mathbf{M}_{DA}, \mathbf{M}_{DB}, \mathbf{M}_{DC}, \mathbf{M}_{DE}, \mathbf{M}_{DF},\\ &\mathbf{M}_{EA}, \mathbf{M}_{EB}, \mathbf{M}_{EC}, \mathbf{M}_{ED}, \mathbf{M}_{EF}, \mathbf{M}_{EI},\\ &\mathbf{M}_{GA}, \mathbf{M}_{HC}, \mathbf{M}_{IE})\end{aligned}$$

of $Q^A, Q^B, Q^C, Q^D, Q^E, Q^F, Q^G, Q^H, Q^I$ be calculated. The assignment in the algorithm OuterConvergence$(\mathbf{x}, \mathbf{M})$ occurs in a manner similar to the assignment in the algorithm InnerAlgorithm$(\mathbf{x}, \mathbf{M})$. Now follows SGD, that reads as

Algorithm 1: Stochastic gradient descent (SGD) update

```
Require: Learning rate schedule ε_1, ε_2, ...
Require: Initial parameter θ
k ← 1
while stopping criterion not met do
    Sample a minibatch of m examples from the training set {x^(1), ..., x^(m)}
      with corresponding targets y^(i)
    Compute gradient estimate ĝ ← (1/m) ∇_θ Σ_i Q(f(x^(i); θ), y^(i))
    Apply update θ ← θ − ε_k ĝ
    k ← k + 1
end
```

Now iterate and deploy!

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